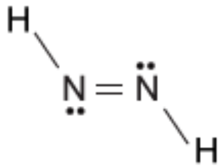
Which of the following molecules has the shortest nitrogen-nitrogen bond?

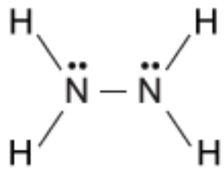
A.

[97%]

B.

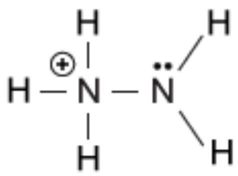


C.



[1%]

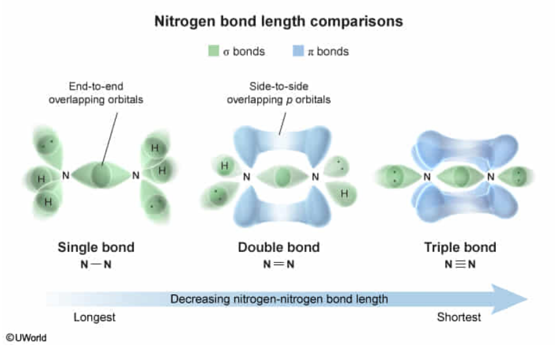
D.



[1%]

Correct
97% answered correctly

Explanation:



Covalent bonds between atoms are formed by sharing electrons, which is made possible through the overlap of the atomic orbitals from each atom. Covalent bonds made by the **end-to-end overlap** of atomic orbitals are called **sigma (σ) bonds** whereas covalent bonds made by the **side-to-side overlap** of *p* orbitals are called **pi (π) bonds**.

Although some atoms participate only in single covalent bonding (by σ bond), other atoms are capable of participating in **multiple bonds** (a double or triple bond) by the addition of one or more π bonds. In this way, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds.

When comparing single, double, and triple covalent bonds involving the *same two types of atoms*, the **bond length** tends to **decrease** (relative to a single bond) with **each added π bond**. This occurs because of the shapes of the orbitals and because of the increased electron density between the atoms participating in multiple bonds.

Of the given molecules, only N_2 has an N≡N triple bond (one σ bond and two π bonds); therefore, N_2 has the shortest of the nitrogen-nitrogen bonds shown.

(Choices B, C, and D) In general, an N–N single bond is longer than an N=N double bond, and an N=N double bond is longer than an N≡N triple bond.

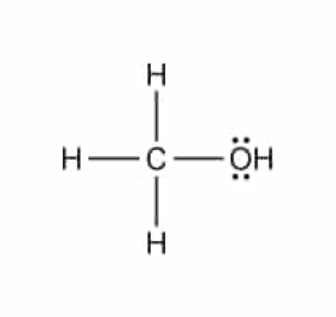
Educational objective:
Covalent bonds are formed by sharing electrons between atoms through the overlap of atomic orbitals in an end-to-end or side-to-side configuration. Multiple bonds formed by *s* and *p* orbitals consist of one σ bond and one or more π bonds, and bond length tends to decrease relative to a single bond with each added π bond.

Time spent:0 sec
Subject:
General Chemistry

QID:402090
System:
Interactions of Chemical Substances

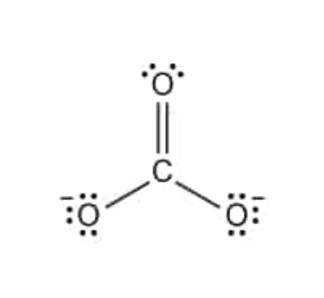
Which of the following molecules has the carbon-oxygen bond with the highest bond dissociation energy?

○ A.



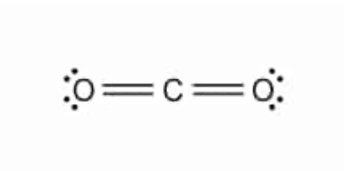
[12%]

○ B.



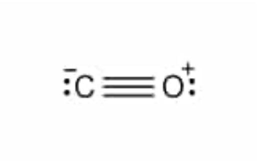
[8%]

○ C.



[8%]

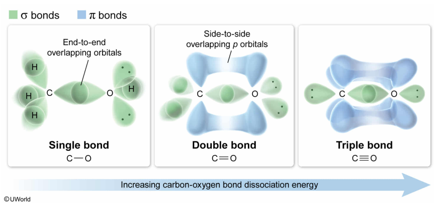
✔ ● D.



[71%]

Correct
72% answered correctly

Explanation:



Covalent bonds between atoms are formed by sharing electrons, which is made possible through the overlap of the atomic orbitals from each atom. Covalent bonds made by the **end-to-end overlap** of atomic orbitals are called **sigma (σ) bonds** whereas covalent bonds made by the **side-to-side overlap** of *p* orbitals are called **pi (π) bonds**.

Although some atoms participate only in single covalent bonding (by σ bond), other atoms are capable of participating in **multiple bonds** (ie, a double or triple bond) by the addition of one or more π bonds. In this way, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds. Comparisons can be made between single, double, and triple covalent bonds provided that the bonds being compared involve the same two types of atoms.

The overall energy of a bond is the sum of the energies of all σ and π bonding contributors. When comparing single, double, and triple covalent bonds involving the same two types of atoms, the overall **bond dissociation energy** (ie, the total energy required to break the σ bond and any π bonds) **increases** (relative to a single bond) with **each additional π bond**.

Out of the molecules given, only carbon monoxide has a C≡O triple bond (one σ bond and two π bonds); therefore, it has the highest bond dissociation energy of the carbon-oxygen bonds shown.

(Choices A, B, & C) In general, a C–O single bond has a lower bond dissociation energy than a C=O double bond, and a C=O double bond has a lower bond dissociation energy than a C≡O triple bond. Adding π bonds increases the number of bonding contributions that must be broken.

Educational objective:
Covalent bonds are formed by sharing electrons between atoms through the overlap of atomic orbitals in an end-to-end or side-to-side configuration. Multiple bonds formed by *s* and *p* orbitals consist of one σ bond and one or more π bonds, and the overall dissociation energy of a bond increases with each additional π bond.

Time spent:0 sec
Subject:
General Chemistry

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System:
Interactions of Chemical Substances